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Examining the Effect of Bat Shape on the Exit Velocity of a Batted Ball

Nava Wolfish[1]

[1] *University of Toronto*

ABSTRACT

The bat-ball collision in baseball occurs on millisecond time scales and is governed by coupled deformation of both the bat and the ball. We model this process using a discretized bat with translational and rotational degrees of freedom per slice, and a nonlinear lossy-spring ball model parameterized by a coefficient of restitution e_0. Instead of expanding the solution in an eigenfunction basis, we directly integrate the coupled equations of motion with RK45 and validate the free-bat dynamics by recovering modal frequencies consistent with the eigenvalue spectrum and prior literature. We then compare standard and Torpedo bat geometries by simulating impacts across barrel locations and evaluating post-collision ball speed and effective coefficient of restitution (eCOR). In a baseline comparison, the Torpedo bat produces a modestly higher and broader peak response near the sweet spot, with a maximum gain of about 1.12 mph in exit velocity. Because those baseline bats differ in density, we perform a second simulation with matched density and similar stiffness to isolate shape effects. Under these controlled conditions, the Torpedo profile still shifts sweet-spot behavior and increases peak exit velocity by about 0.68 mph, while reducing performance outside the sweet-spot region near the tip. These results indicate that bat shape alone can measurably alter collision efficiency and sweet-spot structure, though experimental validation with tracked bat-ball data is required to quantify game-level performance impact.

1. INTRODUCTION

The most crucial moment in baseball occurs within roughly 5 milliseconds. In that time, a batter swings at a pitch and (if they're lucky) makes contact with the ball. The ball flies off the bat with a certain velocity and spin, which will determine if those ≈5 milliseconds will result in a hit, an out (or two), a foul, or even a home run. Physically, these 5 milliseconds contain complex momentum and energy transfers, which depends on the incoming ball, bat swing profile, and the properties of the bat itself. While it's been long known to the baseball world that different bat materials can produce drastically different results, the introduction of the Torpedo bat in 2025 proved that bat shape is also a variable that should be optimized.

The Torpedo bat, designed by Aaron Leanhardt and pioneered with the New York Yankees, refocuses mass towards the "sweet spot" of the bat. For players with a high sweet spot rate, the increase in mass should cause their batted balls to have higher exit velocities and travel further. In theoretical calculations presented on the baseball analysis site Fangraphs, Prof. Alan Nathan predicts that a Torpedo bat should have a wider sweet spot and higher exit velocities farther from the tip compared to a standard bat (A. Nathan 2025). However, physical tests of the Torpedo bat have yet to reproduce Nathan's findings. The Torpedo bat is still so novel that the desired data on exit velocities for batted balls is not yet robust for testing Nathan's theory.

Moreover, the in-game uses of the Torpedo bat do not provide comprehensive control of other factors, such as weather and pitch performance. Given existing use cases, it is impossible to determine whether using the Torpedo bat has any in-game effect on player performance, collision efficiency, or bat speed. In this examination, we ask whether bat shape has a significant effect on the exit velocity of a batted ball. Since the data-driven approach is not yet ready for validation, we instead examine the governing physics equations. We will investigate if theoretical physics simulations enlighten the differences between the Torpedo, standard, and even novel bat shapes.

Existing research on the bat-ball collision is largely provided by Alan Nathan and LL Van Zandt. Van Zandt first proposes that a baseball bat is not a rigid body. Rather, it oscillates or "wiggles" in response to a collision. He models the bat as an accumulation of many small slices (sliced lengthwise, so each cross-section is circular), which are coupled to the adjacent slices via springs. As a result, a small displacement in one bat slice propagates along the

bat. The bat also shears, so each slice can move angularly with respect to the axis of the bat. The frequencies of the propagation depend not only on the stiffness and density of the bat, but also its moment of inertia tensor and each slice's cross-sectional area. Van Zandt solves for the normal modes of the bat and demonstrates how the bat's shape deforms in time as a response to a collision.

Determining the vibrations of the bat is a crucial decomposition. The more energy contained within bat vibration, the less energy transfered to the exit velocity of the batted ball. Baseball players will discuss feeling large vibrations in their hands after a bad swing. They understand that certain locations (normal nodes) along the bat minimize these vibrations and result in better batted ball outcomes.

Nathan elaborates on Van Zandt's work by proposing a different model for the ball and subsequent collision. Nathan models the ball as a nonlinear lossy spring, such that a collision of the ball with a rigid body causes the ball to deform and expand. The non-elastic deformation causes much of the initial kinetic energy of the ball to be lost. Nathan analyzes the energy transfers involved in the collision and ensures that, overall, energy is conserved. Constrastingly, Van Zandt does not account for energy conservation in his model. Nathan argues that energy conservation explains why his model better agrees with observational data compared to Van Zandt.

Since Nathan's paper in 2000, computational methods in physics have improved drastically. The capability of modern computers to integrate complex systems and simulate theoretical results motivate a second look at Nathan's equations. In his original work, Nathan decomposes his equations through the Eigenfuction bases. He finds the vectors and frequencies and theoretically derive how the Eigenfunction amplitudes relate to the forces and velocity of the ball.

Contrastingly, we directly integrate Nathan's equations without need to expand in the Eigenfunction basis. In this methodology, we eliminate an entire step in the integration process. Our end goal is to recreate Nathan's finding through direct integration and extend the model to the Torpedo bat. We then determine if we agree with Nathan's analysis in Fangraphs, or if we can explain discrepancies between his model and observed Torpedo-batted balls.

In this paper...

2. BAT-BALL COLLISION MODEL

2.1. *The Bat*

In accordance with Nathan and Van Zandt, we model the baseball bat discretely. We align the main axis of symmetry of the bat with the z-axis, and slice the bat into N sections of thickness dz perpendicular to the axis (see Figure 1). Each slice of the bat has two degrees of freedom. A slice can displace vertically from the equilibrium position and also angularly with respect to the vertical. We establish coordinates such that ith slice has vertical displacement y_i and angular displacement ϕ_i (see Figure 2). In our examination, we solve for $\Phi_i = \phi_i dz$ to maintain units of length. In total, our bat has $2N$ degrees of freedom.

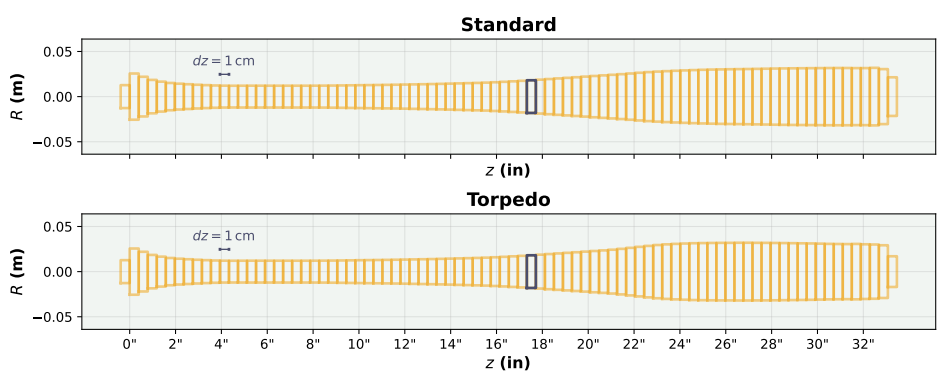

Figure 1. Radial profiles used to define the discretized bat geometry in both models. The top panel shows the standard bat profile and the bottom panel shows the Torpedo bat profile, with each vertical slice corresponding to one numerical element of width $dz = 0.1$ cm. The Torpedo bat redistributes radius, and therefore mass and inertia, toward the mid-barrel region relative to the standard profile.

Crucially, each bat has a measured radial profile $\{R_i\}$, where R_i represents the radius of the ith slice of the bat. The batter's hands grasp the nob, or the 0th slice of the bat. The Nth slice is refered to as the tip. For a solid wooden bat,

each slice then has cross-sectional area $A_i = \pi R_i^2$ and moment of intertia $I_i = \pi R_i^4/4$. Our examination also requires three observed parameters, Young's modulus Y_i, shear modulus S_i, and density ρ_i. A bat is typically constructed from a continuous wooden piece, so we take the reasonable approximation that these parameters are uniform. Henceforth, the subscripts are dropped, and we refer to these parameters as Y, S, ρ.

Van Zandt derives the equations of free motion for a bat from physics first principles. If a single slice is displaced, the neighbouring slices will impose a restoring force to equilibrium. His results align with Timoshenko's beam equations for oscillations of a free beams. He obtains $2N$ coupled 2nd order ordinary differential equations. These are:

$$\ddot{y}_i = \underbrace{\frac{S}{\rho(dz)^2}}_{\text{bat shear}} \left[\underbrace{y_{i+1} + y_{i-1} - 2y_i}_{\text{displacement from neighbours}} + \underbrace{\left(\frac{A_i - A_{i-1}}{A_i}\right)(y_i - y_{i-1} + \Phi_{i-1})}_{\text{correction for non-uniform cross-sections}} + \underbrace{(\Phi_i - \Phi_{i-1})}_{\text{dependence on rotation}} \right] \tag{1}$$

and

$$\begin{aligned} \ddot{\Phi}_i = \underbrace{\frac{Y}{\rho(dz)^2}}_{\text{bat stiffness}} & \left[\underbrace{(\Phi_{i-1} + \Phi_{i+1} - 2\Phi_i)}_{\text{angular displacement from neighbours}} + \underbrace{\left(\frac{I_i - I_{i-1}}{I_i}\right)(\Phi_i - \Phi_{i-1})}_{\text{correction for non-uniform cross-sections}} \right] \\ - \underbrace{\left(\frac{S}{2\rho I_i}\right)}_{\text{bat shear}} & \left[\underbrace{A_i(y_{i+1} - y_{i-1} + \Phi_i + \Phi_{i-1})}_{\text{shear induced torque}} - \underbrace{(A_i - A_{i-1})(y_i - y_{i-1} + \Phi_{i-1})}_{\text{correction for non-uniform cross-sections}} \right]. \end{aligned} \tag{2}$$

Through analysis of these equations, we see that in response to a perturbation we'd expect lighter slices (small A_i) to have a higher maximum displacement compared to more massive slices (high A_i).

In baseball, the knob of the bat is held in the batter's hands. However, the propogation time of a wave along the bat is much slower than the collision time with the ball. On time scales relevant to our problem, we conclude it is appropriate to consider the bat free on both ends. In other words, the first and last slices respond to no external force or torque. They only react to the displacement of internal slices.

The equations of motion are linear and contain no first order terms, so we can rewrite them as:

$$\ddot{\vec{\psi}} = \mathbf{H}\vec{\psi} \tag{3}$$

Where $\mathbf{H}$ is a $2N \times 2N$ non-symmetric matrix, and

$$\vec{\psi} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \\ \Phi_1 \\ \vdots \\ \Phi_N \end{bmatrix}. \tag{4}$$

We can imagine $\mathbf{H}$ as a concatenation of four quadrants, each describing how a $\ddot{\psi}_i$ component depends on a first-order term. The first quadrant relates $\ddot{y}_i$ to y_k, the second relates $\ddot{y}_i$ to Φ_k, the third relates $\ddot{\Phi}_i$ to y_k, and the fourth relates $\ddot{\Phi}_i$ to Φ_k (See Figure 2). Since each slice is only coupled to its neighbours, we expect each quadrant of $\mathbf{H}$ to be tridiagonal. First, we define

$$\Lambda \equiv \frac{S}{\rho(dz)^2} \qquad \Upsilon \equiv \frac{Y}{\rho(dz)^2}. \tag{5}$$

We consider the rows $i \in \{2, 3, ..., N-1\}$. These rows are the central rows of $\mathbf{H}$ and are independent of the boundary conditions imposed on the bat. We also let $j = i + N$ as an index for quadrants 2, 3, and 4. We obtain components of $\mathbf{H}$ from Equations 1 and 2, as outlined in Table 2.

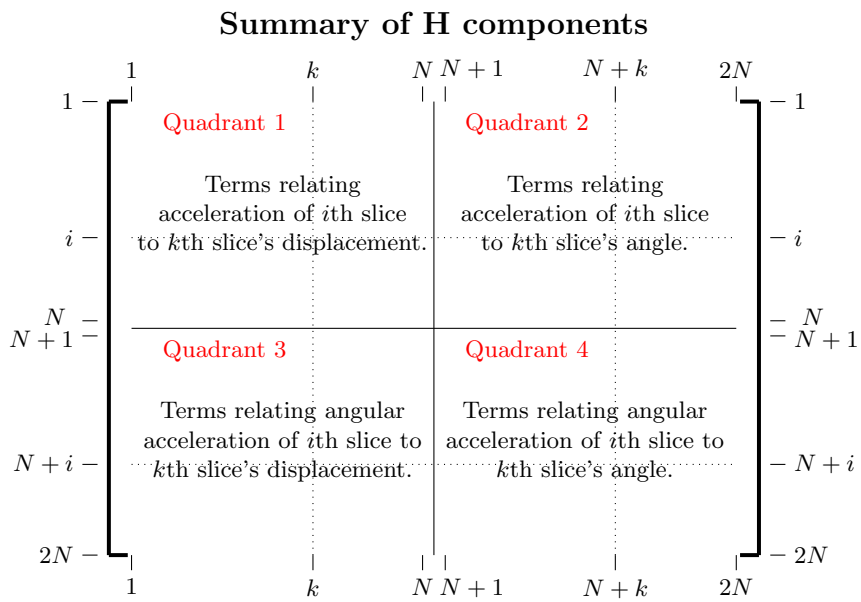

Figure 2. Block-structure schematic for the system matrix **H** in the coupled bat equations of motion. Quadrant 1 maps translational acceleration $\ddot{y}_i$ to displacement y_k, Quadrant 2 maps $\ddot{y}_i$ to angular displacement Φ_k, Quadrant 3 maps angular acceleration $\ddot{\Phi}_i$ to y_k, and Quadrant 4 maps $\ddot{\Phi}_i$ to Φ_k; dotted guides indicate index ranges across the $2N \times 2N$ matrix. This layout shows that translation and rotation are fully coupled and that each block is tridiagonal because each slice interacts only with neighboring slices.

Next, we consider the edges of each quadrant. Invoking our boundary conditions, we obtain the components outlined in Table 3.

All other components of **H** are uniformly zero.

2.2. *Bat Integration and Eigenbasis*

We begin by considering only the bat, without a model for the ball. We take a small perturbation of the kth slice and examine the response wave as it propogates along the bat. To integrate, we employ a Runge-Kutta 4th order integration technique across a uniform time array for a long total time compared to typical bat-ball collision times. Equipped with this solution, we can examinine how the bat vibrations change as a function of affected slice k and the other bat parameters.

We will also examine the normal modes of the bat to justify our results against A. M. Nathan (2000). These modes correspond to standing waves along the length of the bat, with characteristic nodes that do not move as a reponse to a perturbation of the corresponding mode. Since we expect the Eigenvectors of **H** to form a complete and orthonormal basis, all solutions of ψ can be represented as a linear combination of the Eigenfunctions. We can then describe these Eigenfunctions as forming a set of fundamental solutions. The counterpart Eigenvalues describe the frequency at which the bat vibrates for a given Eigenfuction. We expect the Eigenvalues and vectors to take the form

$$\ddot{\psi}_n = -\omega^2 \psi_n \tag{6}$$

where ω is strictly a real number. Positive Eigenvalues indicate that perturbation of a slice would only grow, contrary to our physical understandanding of the system. We do, however, expect two Eigenvectors ψ_n with values of ω near zero that represent rigid body translation and rotation of the bat.

Our final solution will not require us to expand our solution in terms of the Eigenbasis. However, this step is crucial to ensure that our results are quantitatively identical to A. M. Nathan (2000).

We obtain these Eigenfunctions in two ways, and compare to the results in A. M. Nathan (2000). First, we can diagonalize **H** using the appropriate `numpy` functions. This diagonalization should yield the same frequencies and vectors as found by Nathan. We can also take a Fourier transform of the time integrated solution. For a system integrated for a sufficient total time τ, we should recover the same frequencies within the resolution of our transform.

2.3. *The Ball*

According to A. M. Nathan (2000), we model the ball as a lossy spring. We define the coefficient of restitution (COR) e_0, which represents the ratio v_f/v_i for a collision with an immovable rigid body. For example, a ball released from a height h with a COR $e_0 = 0.5$ will rebound from the ground to height $h/4$.

When the ball collides with a rigid body, it contracts by an amount u. Since we are modelling the ball as a spring, we can relate the compression u to the impact force by $F_c(u) = k_1 u^\alpha$, where $\alpha \leq 2$. The ball reaches a maximum compression at $\dot{u} = 0$ and then expands outwards. We have that $u > 0$ represents a compression and $u = 0$ represents the ball at its neutral state. Then $\dot{u} < 0$ is a compressing ball and $\dot{u} > 0$ is an expanding ball. The expansion force curve takes the form $F_e(u) = k_2 u^\beta$, where α and β are related by

$$\beta = \frac{1+\alpha}{e_0^2} - 1. \tag{7}$$

The discrepancies between F_c and F_e describe the energy loss that occurs in the collision. k_2 is determined when the ball reaches maximum compression to retain continuity of the force curve. Figure NUM indicates the difference between the compression and expansion curves. The area between both curves represents the energy lost in the collision and can be parametrized by e_0.

We assume, in a collision, the ball only impacts a single slice of the bat. We can extend to the multi-slice case easily, but the approximation of a single-slice impact is sufficient for our integration. Then, the bat equations of motion extend to

$$\ddot{\vec{\psi}} = \mathbf{H}\vec{\psi} + \vec{F}(t) \tag{8}$$

where

$$\vec{F}(t) = \begin{bmatrix} 0 \\ \vdots \\ F_k(t) \\ \vdots \\ 0 \end{bmatrix}$$

for an impact at slice k.

The physics of the ball are then described by

$$\ddot{y}_b = -\frac{1}{m_b} F(u(t)) \tag{9}$$

$$u(t) = R_b - (y_b(t) - y_k(t)) \tag{10}$$

Where $\dot{u} = 0$ triggers the switch from F_c to F_e, and $u = 0$, $\dot{u} < 0$ triggers the switch from F_e to $F = 0$.

The combination of Equations 8, 9, and 10 form the set of coupled equations we integrate using RK45.

3. RESULTS

We perform the methodology on two bats, whose properties are outlined in Table 4. Figure 1 demonstrates the radial profiles for the standard and Torpedo bats, respectively. Hereon, the superscript (s) denotes the standard bat and (T) denotes the Torpedo bat. Note that the Torpedo bat is 1cm longer than the standard bat, so we have $N^{(s)} = 84$ and $N^{(T)} = 85$.

3.1. *Eigenvectors and Values of* $\mathbf{H}$

We construct the matrix $\mathbf{H}$ for both bats according to the processes outlined in Tables 2 and 2. Before continuing with the time-integration of our system, we examine the Eigenvectors and Eigenvalues of $\mathbf{H}$. We convert ω to f in Hz through $\omega/(2\pi) = f$. The Eigenvalues and corresponding normal frequencies of each bat are listed in 1. As expected, we recover two modes near zero for both bats which represent rigid body translation and rotation. Our values f_n for the standard bat agree with A. M. Nathan (2000), which confirms our construction of $\mathbf{H}$.

The corresponding Eigenmodes for both bats are shown in Figure 3. We only consider the first half of each Eigenvector which correspond to the $\vec{y}$ amplitudes. We observe that, as expected, both bats have nearly identical Eigenvectors.

	Standard		Torpedo	
Mode (n)	Eigenvalue $-(\omega_n^{(s)})^2$	$f_n^{(s)} = \frac{\omega_n^{(s)}}{2\pi}$ (Hz)	Eigenvalue $-(\omega_n^{(s)})^2$	$f_n^{(T)} = \frac{\omega_n^{(T)}}{2\pi}$ (Hz)
Rigid	2.65×10^{-5}	N/A	7.75×10^{-5}	N/A
Rigid	-1.97×10^{-5}	N/A	-3.43×10^{-6}	N/A
1	-1.17×10^{6}	172.38	-8.39×10^{5}	145.77
2	-1.30×10^{7}	574.29	-1.04×10^{7}	515.04
3	-5.48×10^{7}	1178.63	-4.43×10^{7}	1059.49
4	-1.35×10^{8}	1847.03	-1.08×10^{8}	1656.70
5	-2.62×10^{8}	2577.47	-2.11×10^{8}	2309.65

Table 1. Lowest normal-mode eigenvalues and frequencies for the standard and Torpedo bats. Columns list mode index n, eigenvalue $-(\omega_n)^2$, and corresponding frequency $f_n = \omega_n/(2\pi)$ for each bat, and the first two rows are rigid-body modes with near-zero frequency. Both bats share the same qualitative modal structure, while the Torpedo bat exhibits slightly lower frequencies for the first elastic modes.

We also note the positions of the nodes for each modes. If an impact occurs at a node, the corresponding Eigenvector will not be present in the overall solution. Examining the node positions allows us to gauge the impact that hitting a node has on the overall collision efficiency.

3.2. *Free Bat Oscillations*

We first allow our bats to oscillate freely without considering the model for the ball. We set the initial conditions for both bats to zero displacement and velocity across all slices. Then, we displace the 75th slice (at $z = 0.75$ m) by 0.1 mm. We use a time step of $dt = 5$ μs and integrate using RK45 for a total time of $\tau = 50$ ms. A heatmap of the results is shown in Figure 4. We observe the regions of each bat with smaller cross-section A_i have higher rotational and translational displacement. We also notice more prominent rotational displacement for the Torpedo bat compared to the standard bat. Nevertheless, the rotational solution is much smaller than the translational solution. This fact confirms our focus on the $\{y_i\}$ displacements rather than $\{\Phi_i\}$.

3.2.1. *Fourier Decomposition*

We take a Fourier decomposition in time of the $\{y_i\}$ solutions for both the Torpedo and standard bats. We then average the frequency spectrum across all slices, such that we obtain an average amplitude-frequency array for both bats. Since we have $\tau = 50$ ms, our Fourier transform will have a frequency resolution of $\Delta f = 1/\tau = 20$ Hz. We find peaks in each spectrum using a centre-of-mass centroid algorithm. Since we impacted a node of the 3rd mode, we only expect to recover f_1, f_2 for both bats. We obtain $f_1^{(s)} = 178.5$ Hz and $f_2^{(s)} = 576.6$ Hz for the standard bat. Both frequencies are within 20 Hz of their expected values from Table 1. We obtain $f_1^{(T)} = 144.0$ Hz and $f_2^{(T)} = 516.9$ Hz. These values are also within 20 Hz of their expected values.

Since the time-integration converges reliably and we can successfuly recover the expected frequencies from the time-integrated solutions for both bats, we continue to involve the ball in our simulations.

3.3. *Simulation #1*

According to MLB regulations and lab measurements conducted by Prof. Rod Cross at the University of Sydney, we select a baseball with the parameters outlined in Table 5.

We perform the integration of Equations 8, 9, and 10 with the proper definition of F for the current values of u and $\dot{u}$. When the integrator triggers $\dot{u} = 0$, we calculate β using Equation 7 and switch from $F_c \to F_e$. After expansion, when $u = 0$ and $t > 0$, $F \to 0$.

We impact the bat with a ball travelling at $v_{b,0}(t = 0) = 44.704$ m/s or 100 mph, which is a typical differential velocity between the bat and ball for baseball contexts. We begin with $y_b(t = 0) = R_b$, such that we isolate the moment that the ball first makes contact with the bat. Both bats start at rest with $y_n(0) = 0$ for all n. We simulate the bat-ball collision across the axis of the bat, for each slice $45 \leq i \leq 83$ and each bat (s), (T).

After the bat and ball separate, the ball continues with constant velocity $v_{b,f}$. Since we are only concerned with the immediate collision, we do not consider the factors that affect ball flight after separation. We take the value of $v_{b,f}$ across all collisions and compare both bats, shown in Figure 6. Importantly, we expect the ratio $v_{b,f}/v_{b,0} \leq e_0$.

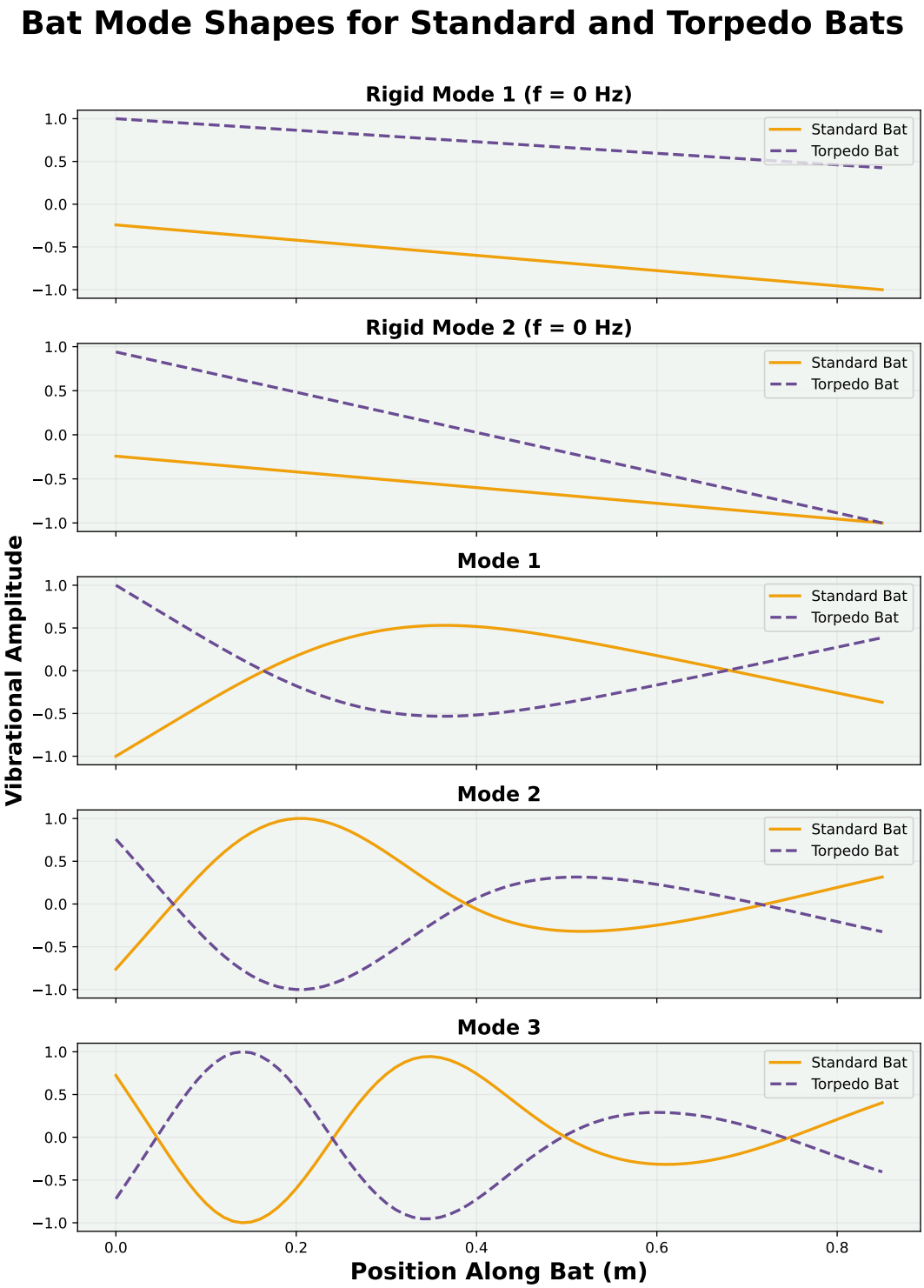

Figure 3. First mode shapes (displacement components) of the standard and Torpedo bats. Each curve corresponds to a normal mode as a function of bat position, and node locations occur where modal displacement crosses zero. The Torpedo modes are in dashed-purple and the standard bat modes are in solid-yellow. The two bats have very similar mode-shape geometry, so the dominant differences in collision response arise from subtle shifts in mode frequency and local mass distribution rather than wholesale changes in modal form.

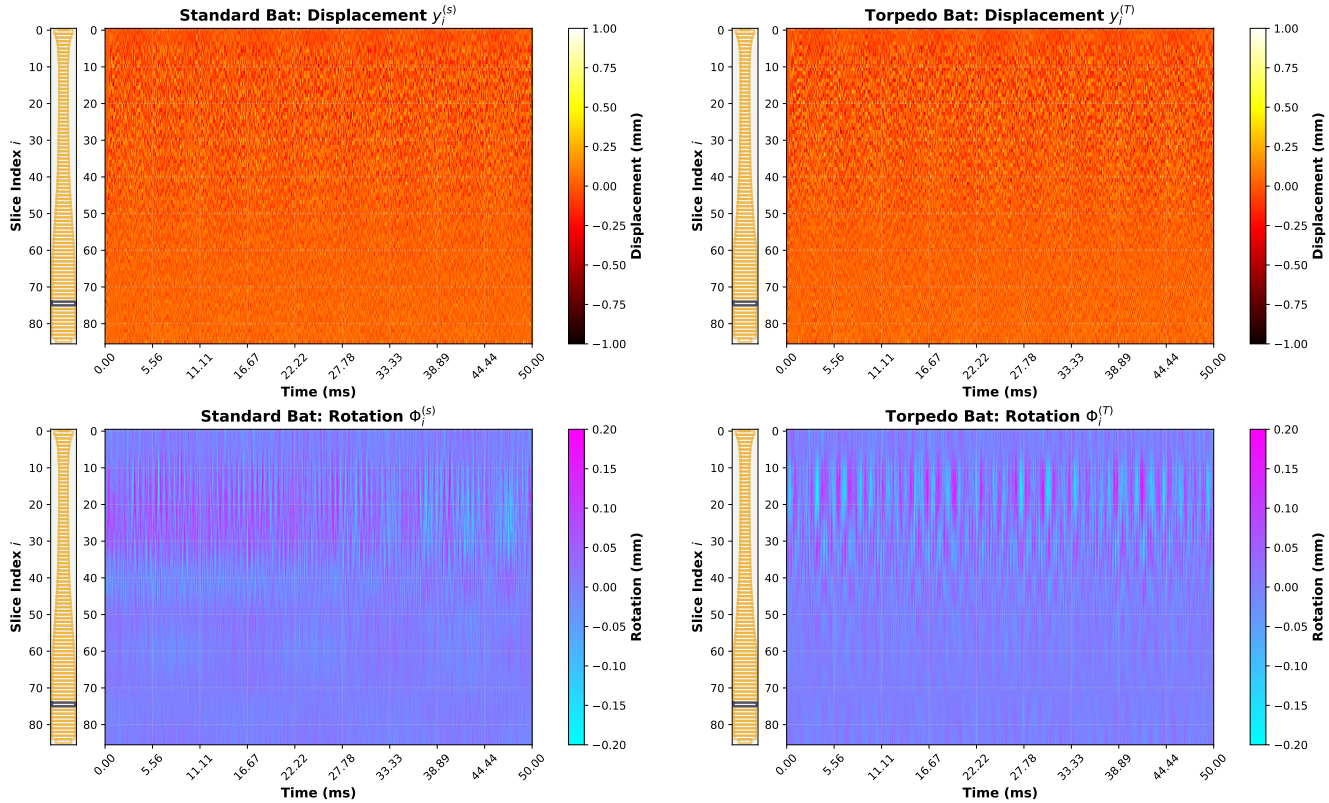

Figure 4. Time-evolution heatmaps of free oscillations after a localized perturbation at slice $k = 75$. Color indicates displacement magnitude and sign as a function of slice position and time, and the panels compare standard and Torpedo bats for translational and rotational responses. Larger amplitudes occur in thinner sections, and the Torpedo geometry shows stronger localized rotational response while translational motion remains dominant overall.

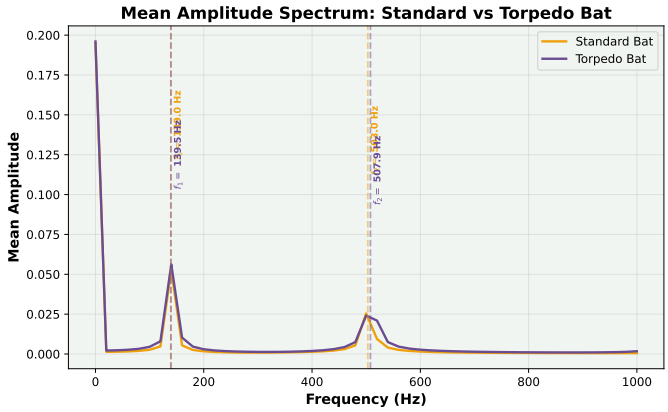

Figure 5. Frequency spectra from Fourier transforms of free-oscillation solutions. Amplitude is plotted versus frequency for the standard and Torpedo bats, and spectral peaks identify the excited normal modes. The purple line is the Torpedo amplitude spectrum and the yellow line is the standard amplitude spectrum. The recovered peak frequencies align with eigenvalue-based mode frequencies within the transform resolution, validating the time-integration approach.

Some of the energy from the collision is transfered to the vibration of the bats, so the collision should not yield the maximum allowed exit velocity for the ball. We let $v_{b,f}/v_{b,0} = e_e$ be the effective COR for each collision.

For both bats, the exit velocity peaks at $z = 67$ cm, around 6.6 inches from the tip of the bat. At this point the standard bat has an eCOR of 0.264177 and the Torpedo bat has a eCOR of 0.275214. The exit velocties and eCORs taper down on either side of the peak. This observed maximum is consistent with findings in R. Cross (1998). The $\approx$

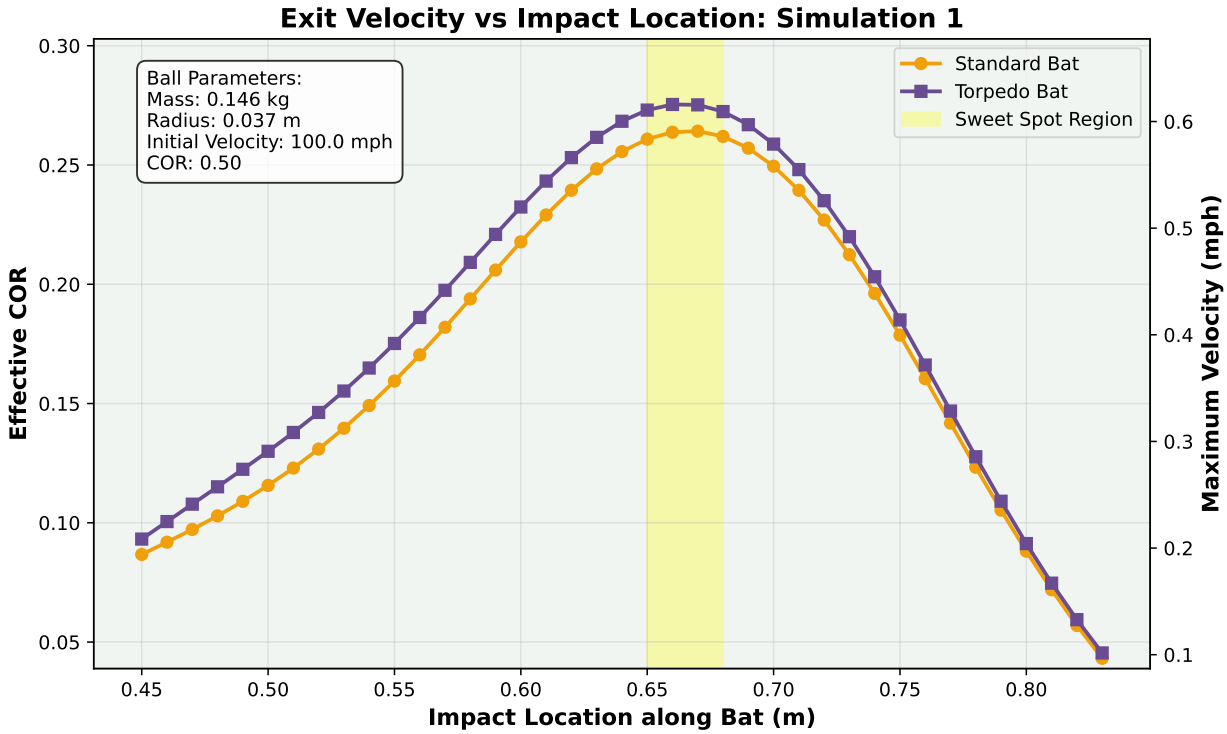

Figure 6. Simulation 1 impact-location dependence of post-collision ball speed (or effective COR) for standard and Torpedo bats. The horizontal axis is impact position along the bat, and each curve represents one bat profile under the Simulation 1 material parameters. The Torpedo bat is plotted in purple and the standard bat is plotted in yellow. Both bats peak near the sweet spot, but the Torpedo bat produces a slightly higher and broader peak response in this setup.

3 cm region where the velocity peaks is known as the "sweet spot", and is typically found 18 cm (7 inches) from the tip of the bat.

We observe the Torpedo bat has marginally higher exit velocity for all collisions, however the difference in performance becomes stark near the sweet spot of the bat. At peak, the Torpedo bat gives an extra 0.5 m/s = 1.12 mph of velocity to the ball. We also observe the Torpedo bat produces a broader sweet spot, yielding higher eCOR accross a wider range of impact locations. This result is consistent with Nathan's predictions in Fangraphs.

We notice, however, that the Torpedo bat has an additional 41 kg/m^3. This increase in density resists vibration and causes more energy to be transfered to the ball. As a result, we cannot definitively conclude our results are attributed to the difference in shape.

3.4. *Simulation #2*

We repeat the same process as in Simulation #1 with two new bats. The parameters for these bats are outlined in Table 4. Notably, both bats have the same density and similar stiffness. These bats isolate nearly all parameters but bat shape. We analyze the results in Figure 7.

In this simulation, we find the Torpedo bat has a sweet spot shifted inwards compared to the standard bat. While the Torpedo bat has peak effective COR of 0.275 at $z = 67$ cm, the standard bat has peak effective COR of 0.270 at $z = 68$ cm. Again, we observe a broader sweet spot for the Torpedo bat and large exit velocity gain between $z = 60$ cm and $z = 67$ cm. However, on the exterior of the sweet spot, the Torpedo bat has lower effective COR compared to the standard bat.

Since this simulation controls for density and stiffness, the differences between both bats is wholly attributed to bat shape.

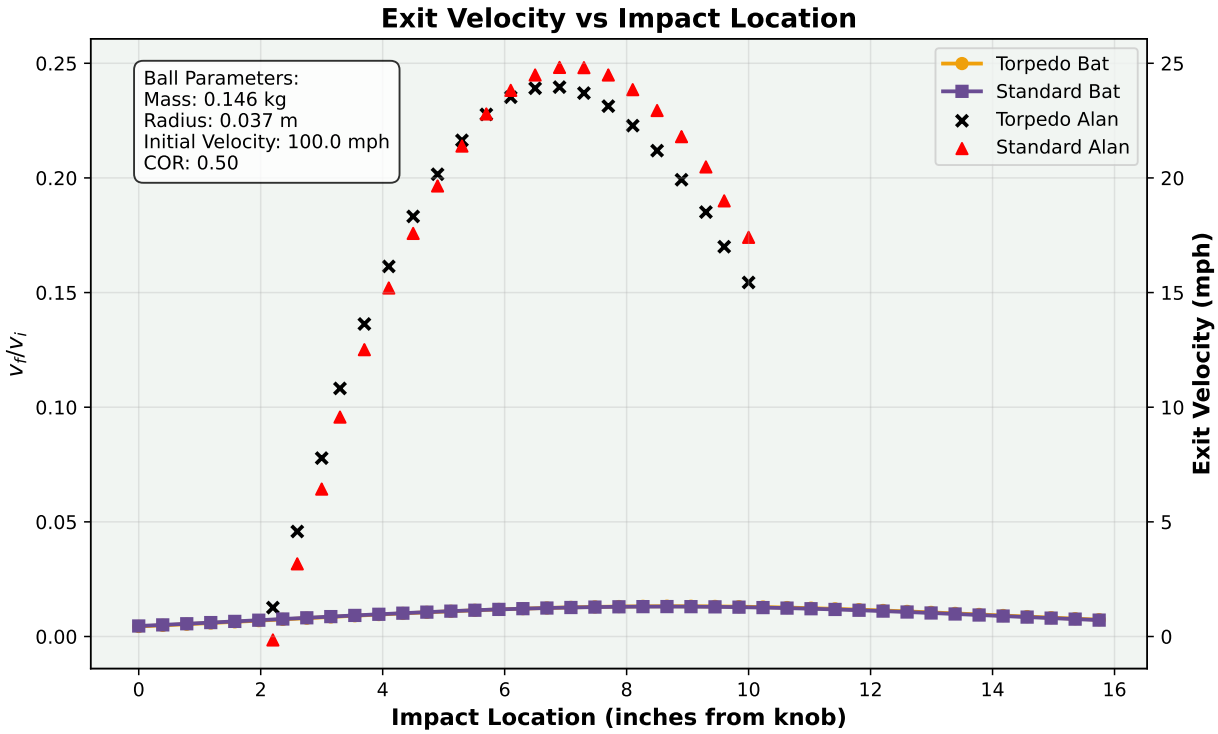

Figure 7. Simulation 2 impact-location dependence of post-collision performance with matched-density bats. Curves show standard versus Torpedo results under the Simulation 2 parameter set, where density and stiffness are controlled to isolate geometry effects. The Torpedo bat is plotted in purple and the standard bat is plotted in yellow. Differences persist after parameter matching, indicating that bat shape alone can shift sweet-spot location and local collision efficiency.

4. DISCUSSION

In this study, we used computational techniques to examine the quantitative differences between batted balls for standard and Torpedo bats. We analyzed the degrees of freedom of the bats and found the normal modes for free vibrations. Our results were consistent with A. M. Nathan (2000). We noticed both bats had nearly identical Eigenmodes, however the Torpedo bat had slightly lower normal frequencies.

We then modelled the ball as a lossy spring with coefficient of restitution e_0. We determined the force present between the bat and the ball and related this force to the ball's position and compression. We implemented these forces into our system of equations and created an integration model using RK45.

First, we simulated the collision with a 100 mph ball for Nathan's standard bat and a Torpedo bat. We noticed the Torpedo bat had marginally higher exit velocities, notably giving the ball 1.12 mph more than the standard bat at peak exit velocity. However, the bats in this simulation had significantly different densities, so we could not draw significant conclusions about the effect of bat shape. We then repeated the simulation for bats of identical densities. We found that controlling for density only added 0.68 mph to the ball's exit velocity at peak.

Nevertheless, this increase in speed is considerable in baseball. With an additional 0.68 mph, many warning-track (very close to the wall) fly balls gain an extra few feet and become home runs. Giving players this advantage is hugely important to MLB teams and is impactful for players with high fly ball rates that fall short of powerful superstars.

We also noticed the Torpedo bat had a broader sweet spot, particularly towards knob. This result implies the Torpedo bat may be best suited for players that are less consistent with their impact location, particularly those that miss the sweet spot on the inside. However, players that hit the ball towards the tip should continue using the standard bat, as the standard bat outperforms the Torpedo bat outside the sweet spot.

This study confirms that the shape of a baseball bat contributes meaningfully to a collision with a ball. Further study into bat shape, energy conservation in the collision, and ball flight is required to determine how we can alter bat shape to create better baseball players.

Furthermore, this study does not leverage any observational data on incoming and exit velocities of batted balls. Before we can conclusively state that the Torpedo bat yields faster balls, we must analyze whether our theoretical results agree with lab data.

APPENDIX

A. MATRIX COMPONENT TABLES

	Quadrant 1	**Quadrant 2**
Diagonal	relevant indices $\{(i,i)\}$	relevant indices $\{(i,j)\}$
Upper	$H_{i,i+1} = \Lambda$	$H_{i,j+1} = \frac{\Lambda}{2}$
Main	$H_{i,i} = -\Lambda\left(1 + \frac{A_{i-1}}{A_i}\right)$	$H_{i,j} = \frac{\Lambda}{2}\left(1 - \frac{A_{i-1}}{A_i}\right)$
Lower	$H_{i,i-1} = \Lambda\frac{A_{i-1}}{A_i}$	$H_{i,j-1} = -\Lambda\frac{A_{i-1}}{2A_i}$
	Quadrant 3	**Quadrant 4**
Diagonal	relevant indices $\{(j,i)\}$	relevant indices $\{(j,j)\}$
Upper	$H_{j,i+1} = -\Lambda\frac{A_i(dz)^2}{2I_i}$	$H_{j,j+1} = \Upsilon - \Lambda\frac{A_i(dz^2)}{4I_i}$
Main	$H_{j,i} = \Lambda\frac{A_i(dz)^2}{2I_i}\left(1 - \frac{A_{i-1}}{A_i}\right)$	$H_{j,j} = -\Upsilon\left(1 + \frac{I_{i-1}}{I_i}\right) - \Lambda\frac{A_i(dz)^2}{4I_i}\left(1 + \frac{A_{i-1}}{A_i}\right)$
Lower	$H_{j,i-1} = \Lambda\frac{A_{i-1}(dz)^2}{2I_i}$	$H_{i,j-1} = -\Upsilon\left(\frac{I_{i-1}}{I_i}\right) - \Lambda\frac{A_i(dz)^2}{4I_i}$

Table 2. Central (interior-slice) coefficient definitions for the system matrix $\mathbf{H}$. Rows are split by upper, main, and lower diagonals and columns are split by Quadrants 1–4, with each expression giving a nonzero coefficient for interior indices $i \in \{2, \ldots, N-1\}$. All interior couplings are nearest-neighbor and parameterized by local area and inertia ratios, producing a sparse tridiagonal block structure.

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	Quadrant 1	**Quadrant 2**
Boundary	relevant indices $\{(i,i)\}$	relevant indices $\{(i,j)\}$
Knob	$H_{1,1} = -\Lambda$ $H_{1,2} = \Lambda$	$H_{1,N+1} = \frac{\Lambda}{2}$ $H_{1,N+2} = \frac{\Lambda}{2}$
Tip	$H_{N,N-1} = \Lambda \frac{A_{N-1}}{A_N}$ $H_{N,N} = -\Lambda \frac{A_{N-1}}{A_N}$	$H_{N,2N-1} = -\Lambda \frac{A_{N-1}}{2A_N}$ $H_{N,2N} = -\Lambda \frac{A_{N-1}}{2A_N}$
	Quadrant 3	**Quadrant 4**
Boundary	relevant indices $\{(j,i)\}$	relevant indices $\{(j,j)\}$
Knob	$H_{N+1,1} = -\Lambda \frac{A_N (dz)^2}{4I_N}$	$H_{j,j+1} = \Upsilon - \Lambda \frac{A_i (dz^2)}{4I_i}$
Main	$H_{j,i} = \Lambda \frac{A_i (dz)^2}{2I_i} \left(1 - \frac{A_{i-1}}{A_i}\right)$	$H_{j,j} = -\Upsilon \left(1 + \frac{I_{i-1}}{I_i}\right) - \Lambda \frac{A_i (dz)^2}{4I_i} \left(1 + \frac{A_{i-1}}{A_i}\right)$

Table 3. Boundary (knob and tip) coefficient definitions for the system matrix **H**. Entries list nonzero matrix elements for boundary rows in each quadrant, replacing interior formulas where free-end conditions modify the coefficients. These boundary corrections are essential to enforce physically consistent free-end force and torque behavior in the discretized bat model.

	Simulation 1		**Simulation 2**		
Property	**Standard Bat**	**Torpedo Bat**	**Standard**	**Torpedo**	Units
Bat Length	0.84	0.85	0.84	0.84	m
Density ρ	649	690	690	690	kg/m^3
Shear Modulus S	1.05	0.9	1.0	0.9	$\times 10^9$ N/m^2
Young's Modulus Y	1.814	1.650	1.64	1.65	$\times 10^{10}$ N/m^2
Slice Width dz	0.01	0.01	0.01	0.01	m
Num. Slices N	84	85	85	85	slices

Table 4. Input bat properties used across the two collision simulations. Columns report standard and Torpedo values for Simulation 1 and Simulation 2, while rows list geometric and material parameters (L, ρ, S, Y, dz, and N). Simulation 1 compares realistic but materially different bats, whereas Simulation 2 controls density and stiffness to isolate the effect of bat shape.

Parameter	Symbol	Value	Unit
Mass	m_b	0.146	kg
Radius	R_b	0.0366	m
Initial Velocity	$v_{b,0}$	44.704	m/s
Coefficient of Restitution	e_0	0.50	dimensionless
Compression Constant	k_1	0.714×10^7	$\mathrm{N/m^{\alpha}}$
Compression Exponent	α	1.36	dimensionless

Table 5. Baseball model parameters used in all integrations. Rows list mass, radius, initial relative impact speed, coefficient of restitution e_0, compression constant k_1, and compression exponent α for the nonlinear lossy-spring ball model. These fixed ball parameters define the compression-expansion force law and allow direct comparison of bat-dependent outcomes.